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3.2.3 Classical Gram-Schmidt algorithm

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Video

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3.2.3
CGS algorithm in FLAME notation

Robert van de Geijn
Maggie Myers

0:00 / 0:00

2.0x

Start of transcript.

Skip to the end.

The way we've described the Gram Schmidt Orthogonalization process, and of course since that computes the QR factorization, therefore how to compute the QR factorization, involves a lot of indices. And our experience is that

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Reading assignment

1/1 point (graded)



Read Unit 3.2.3 of the notes. [\[LINK\]](#)

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Homework 3.2.3.1

1/1 point (graded)
Implement the algorithm

$[Q, R] = \text{CGS_QR}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_L & A_R \end{array} \right), Q \rightarrow \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right), R \rightarrow \left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right)$

$A_L \text{ and } Q_L \text{ has } 0 \text{ columns and } R_{TL} \text{ is } 0 \times 0$

$\text{while } n(A_L) < n(A)$

$\left(\begin{array}{c|c} A_L & A_R \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_0 & a_1 & A_2 \end{array} \right), \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right) \rightarrow \left(\begin{array}{c|cc} Q_0 & q_1 & Q_2 \end{array} \right),$

$\left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ \hline 0 & 0 & R_{22} \end{array} \right)$

$r_{01} := Q_0^T a_1$

$a_1^\perp := a_1 - Q_0 r_{01}$

$\rho_{11} := \|a_1^\perp\|_2$

$q_1 := a_1^\perp / \rho_{11}$

$\left(\begin{array}{c|c} A_L & A_R \end{array} \right) \leftarrow \left(\begin{array}{c|cc} A_0 & a_1 & A_2 \end{array} \right), \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right) \leftarrow \left(\begin{array}{c|cc} Q_0 & q_1 & Q_2 \end{array} \right),$

$\left(\begin{array}{c|c} R_{TL} & R_{TR} \\ \hline 0 & R_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} R_{00} & r_{01} & R_{02} \\ \hline 0 & \rho_{11} & r_{12}^T \\ \hline 0 & 0 & R_{22} \end{array} \right)$

endwhile

as

```
function [ Q, R ] = CGS_QR( A )
```

by completing the code in [Assignments/Week03/matlab/CGS_QR.m](#).
Input is an $m \times n$ matrix A . Output is the matrix Q and the upper triangular matrix R . You may want to use [Assignments/Week03/matlab/test_CGS_QR.m](#) to check your implementation.

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